Coordinate Geometry

Section 1: Slopes, Intercepts & Lines

1 What is the slope of the line perpendicular to the line with equation $3x - 4y = 12$?

a. $\frac{4}{3}$

b. $-\frac{4}{3}$

c. $\frac{3}{4}$

d. $-\frac{3}{4}$

2 A line passes through the points $(2, -3)$ and $(6, 5)$. What is the y-intercept of this line?

a. -7

b. -5

c. -3

d. 1

3 Which of the following points lies on the line that passes through $(1, 4)$ and has a slope of -3?

a. $(3, -2)$

b. $(2, 2)$

c. $(4, -4)$

d. $(0, 6)$

Section 2: Midpoints & Distance Formula

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 What is the distance between the points $A(-3, 2)$ and $B(5, -4)$ in the standard (x, y) coordinate plane?

a. 8

b. 10

c. 12

d. 14

5 The midpoint of line segment $\overline{CD}$ is $M(4, -1)$. If point C has coordinates $(1, 5)$, what are the coordinates of point D?

a. $(7, -7)$

b. $(6, -6)$

c. $(5, -3)$

d. $(7, -6)$

6 What is the equation of the perpendicular bisector of the segment connecting $(0, 2)$ and $(4, 6)$?

a. $y = -x + 6$

b. $y = -x + 4$

c. $y = x + 2$

d. $y = -2x + 8$

Section 3: Circles in the Coordinate Plane

7 What are the center and radius of the circle given by $(x - 5)^2 + (y + 2)^2 = 49$?

a. Center $(5, -2)$, radius 7

b. Center $(-5, 2)$, radius 7

c. Center $(5, -2)$, radius 49

d. Center $(-5, 2)$, radius 49

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 A circle has the equation $x^2 + y^2 - 6x + 8y = 0$. What is the radius of this circle?

a. 4

b. 5

c. 6

d. 10

9 Which of the following lines is tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$?

a. $3x + 4y = 25$

b. $4x + 3y = 25$

c. $3x - 4y = 25$

d. $4x - 3y = 0$

Section 4: Parabolas & Conic Forms

10 What is the vertex of the parabola defined by the equation $y = 2x^2 - 8x + 11$?

a. $(2, 3)$

b. $(2, 5)$

c. $(-2, 35)$

d. $(4, 11)$

11 What is the equation of the axis of symmetry for the quadratic function $y = -3(x + 4)^2 - 7$?

a. $x = -4$

b. $x = 4$

c. $y = -7$

d. $x = -7$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 At how many points do the graphs of $y = x^2 - 4x + 5$ and $y = 2x - 4$ intersect?

a. 0

b. 1

c. 2

d. Infinitely many

© ClipSAT: By Mr. Mohamed Abdallah +966597688647